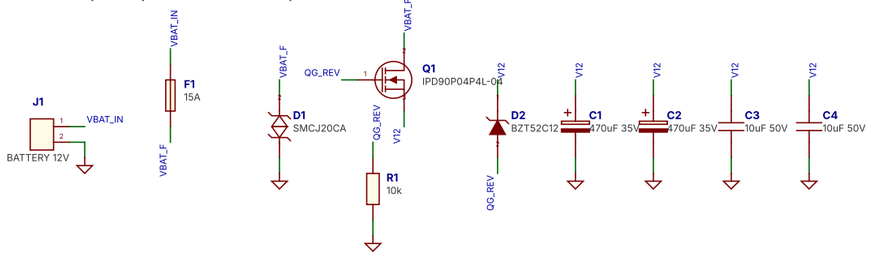
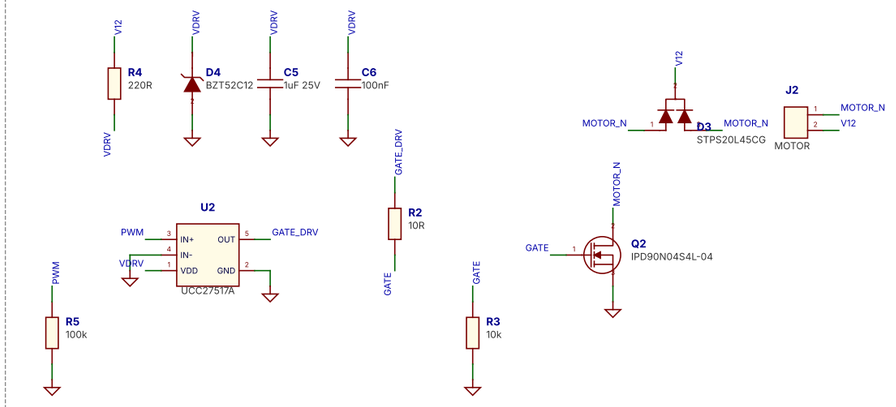
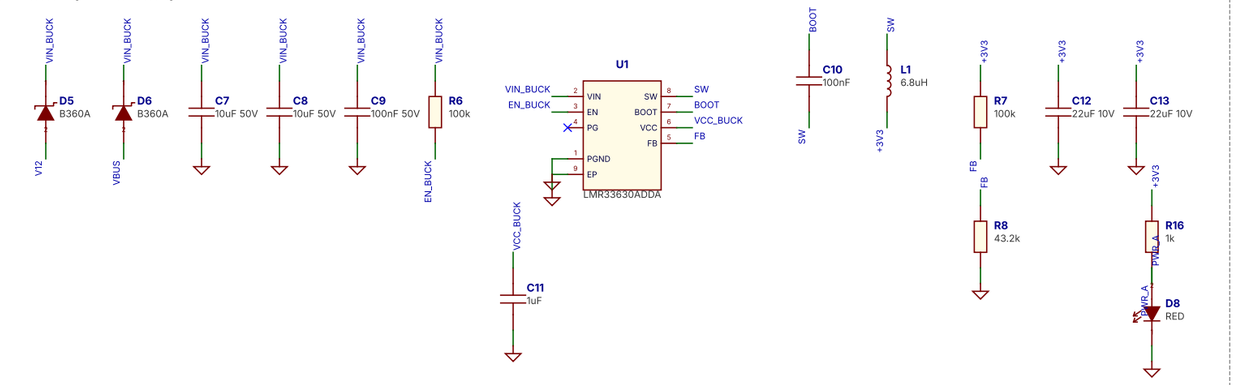
POWER INPUT, FUSE, LOAD-DUMP TVS, REVERSE-POLARITY PROTECTION



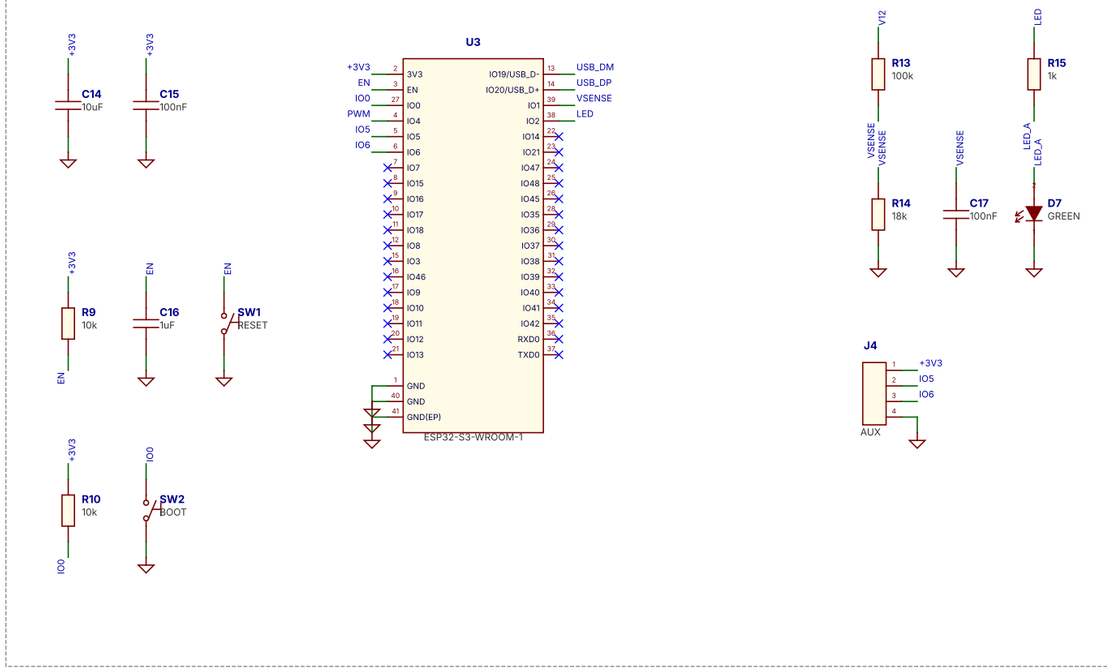
MOTOR SWITCH: GATE DRIVER + LOW-SIDE N-MOSFET + FREEWHEEL DIODE



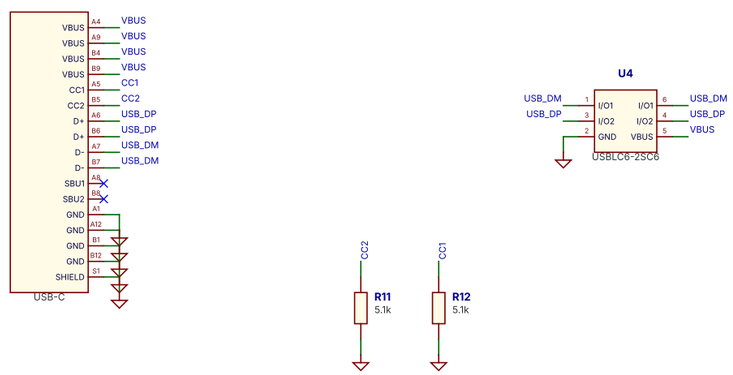
3.3 V BUCK (12 V or USB)



ESP32-S3 MODULE, RESET/BOOT, BATTERY SENSE, LEDS, AUX



USB-C (PROGRAMMING / DEBUG)



Motor: J2 pin 2 = +12 V (always on), pin 1 = MOTOR- (switched to GND by Q2).
PWM from ESP32 GPIO4, 20 kHz. R5 holds the gate driver input low while the ESP32 boots.
Battery sense: VSENSE = VBAT x 18k / 118k → 14.4 V reads 2.20 V on GPIO1.
Q1 + D2: reverse-battery protection (P-MOSFET, gate clamped to 12 V).
D1 SMCJ20CA clamps load-dump spikes; everything on the 12 V side is rated >= 35 V.
Rated for motors up to 10 A continuous (15 A fuse). Use 2 oz copper.

H1 M3 H2 M3 H3 M3 H4 M3

ESP32-S3 12 V Motor PWM Controller

Rev 1.0 2026-10-26 Sheet 1/1 A3

12 V in (11-14.4 V) → PWM low-side motor switch, up to 10 A

Labels with the same name are connected.